Shiv Nadar University, Chennai
Department of Information Technology
25IT1731 Computational Thinking and Problem Solving with Python

Ex 6 – Dictionary

Part A

1	Create a dictionary named student containing the details: Name – Arun, Age – 20, Department – IT, and Mark – 85. a) Display the entire dictionary. b) Display the student's name and mark. c) Access the age using its key. d) Use get() to access the key "Grade". Display a suitable message if the key does not exist.
2	Consider the following dictionary: student = {"Name": "Priya", "Age": 19, "Department": "CSE", "Mark": 92} a) Display all keys and values separately. b) Display all key-value pairs using items(). c) Use a for loop to display each entry in the format Name : Priya.
3	A bookstore maintains the following dictionary: books = {"Python": 450, "Java": 500, "C": 300} a) Add C++ with a price of ₹400. b) Update the price of Python to ₹480. c) Remove C using pop(). d) Display the number of books and check whether Java is present. e) Use update() to add SQL priced at ₹350. f) Create a copy of the dictionary, clear the copied dictionary, and display it.
4	A music application stores song titles and durations in two lists: songs = ["Shape of You", "Believer", "Perfect", "Memories"] duration = [4.2, 3.4, 4.4, 3.1] a) Combine the lists into a dictionary using zip(). b) Display each song and its duration. c) Find the longest song. d) Display songs shorter than four minutes. e) Calculate the total playlist duration.
5	Two supermarket branches record the quantities of products sold: branch1={"Rice":20,"Sugar":15,"Oil":10} branch2 = {"Rice": 12, "Sugar": 8, "Wheat": 18} a) Combine both dictionaries into one. b) If a product appears in both dictionaries, add its quantities. c) Display the final product-wise quantities.
6	An online shopping website wants to identify frequently occurring words in customer reviews. review = "good product good quality excellent product good quality" a) Count the frequency of each word. b) Treat uppercase and lowercase versions of a word as the same.

	c) Display only the words occurring more than once.
7	Consider the following dictionary: products = {"Laptop": 55000, "Mouse": 700, "Keyboard": 1500, "Monitor": 12000, "Webcam": 2500} a) Sort the dictionary by product name in ascending order. b) Sort it by product name in descending order. c) Sort it by price in ascending order. d) Sort it by price in descending order. e) Display the most expensive and least expensive products.

8	A bank maintains account details using the following dictionary:
	accounts = {
	"1001": {"name": "Arun", "balance": 5000},
	"1002": {"name": "Priya", "balance": 8000},
	"1003": {"name": "Kumar", "balance": 3000}
	}
	a) Accept an account number and display the account holder's name and balance.
	b) Deposit an amount into an account after checking that the amount is positive.
	c) Withdraw an amount only if sufficient balance is available.
	d) Display all account holders and their current balances.
	e) Calculate the total amount maintained across all bank accounts.

9	A college stores student details and assessment marks:
	students = {
	"101": {"Name": "Arun", "Python": {"CAT1": 80, "CAT2": 85}, "Java": {"CAT1": 75, "CAT2": 90} },
	"102": {"Name": "Priya","Python": {"CAT1": 92, "CAT2": 95}, "Java": {"CAT1": 88, "CAT2": 91} }
	}
	a) Display all student details.
	b) Display the Python CAT1 mark of student 101.
	c) Update Priya's Java CAT2 mark to 96.
	d) Add a new student with registration number 103 and assessment marks.
	e) Calculate the total marks obtained by each student across all four assessments.
	f) Display the student who obtained the highest total.

10	Consider the following dictionary:
	student_marks = {
	"Arun": 85,
	"Priya": 92,
	"Kavin": 85,
	"Meena": 78,
	"Divya": 92,
	"Rahul": 88
	}
	Write a Python program to accept a mark from the user and perform a reverse lookup to display the names of all students who obtained that mark. If no student has the given mark, display an appropriate message.

Part B

11	A weather station records daily weather measurements for different cities.
	weather = {
	"Chennai": {"temperature": 34.5, "humidity": 70, "rainfall": 12.5},
	"Coimbatore": {"temperature": 31.0, "humidity": 65, "rainfall": 8.0},
	"Madurai": {"temperature": 36.2, "humidity": 60, "rainfall": 4.5},
	"Ooty": {"temperature": 22.5, "humidity": 80, "rainfall": 25.0}
	}
	Write a Python program to perform the following operations:
	1. Display weather details for all cities.
	2. Add weather information for a new city.
	3. Update the temperature, humidity, or rainfall for a city.
	4. Display cities where the temperature exceeds 33°C.
	5. Identify the city with the highest temperature.
	6. Identify the city with the highest rainfall.
	7. Calculate the average temperature across all cities.
	8. Sort cities by temperature in ascending order.
	9. Display cities with humidity above 70%.
	10. Calculate the total recorded rainfall across all cities.

12	A city festival committee maintains information about different festivals using the following :
	festivals = {
	"Pongal": {"activities": ["Kolam", "Folk Dance", "Cooking Contest"],
	"days": 3, "budget": 25000.0, "location": "Chennai"},
	"Music Festival": {"activities": ["Classical Music", "Light Music", "Dance"],
	"days": 2, "budget": 18000.0, "location": "Coimbatore"},
	"Food Festival": {"activities": ["Cooking Show", "Food Stalls", "Tasting Event"],
	"days": 4, "budget": 30000.0, "location": "Madurai"},
	"Book Festival": {"activities": ["Book Exhibition", "Author Talk", "Quiz"],
	"days": 2, "budget": 12000.0, "location": "Chennai"}
	}
	1. Display all festivals and their details.
	2. Add a new festival with its activities, duration, budget, and location.
	3. Update the budget and duration of an existing festival.
	4. Display festivals whose budget is below an amount entered by the user.
	5. Recommend festivals that have at least three activities and last no more than three days.
	6. Identify the festivals with the highest and lowest budgets.
	7. Sort festivals by budget in ascending order.
	8. Calculate the total budget of all festivals.
	9. Add a new activity to an existing festival.
	10. Display all festivals conducted in Chennai.

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